

TEMASEK JUNIOR COLLEGE, SINGAPORE

JC 2 and TA 4

Preliminary Examination 2008

Higher 1

MATHEMATICS

8863

8 September 2008

Additional Materials: Answer paper

3 hours

List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your *Civics Group* and *Name* on all the work that you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This document consists of 7 printed pages and 1 blank page.



Section A: Pure Mathematics [40 marks]

- 1** Find $\int_a^0 (e^{-2x} + x^2)(e^{-2x} - x^2) dx$ in terms of a , where $a < 0$. [3]

Hence find a if $\int_a^0 (e^{-2x} + x^2)(e^{-2x} - x^2) dx = 1$, leaving your answer correct to 3 decimal places. [1]

- 2** The functions f and g are defined below:

$$f : x \mapsto e^{x^2+x}, \quad x \in \mathbb{R}, \quad x \geq -\frac{1}{2},$$

$$g : x \mapsto \frac{1}{\sqrt{x}}, \quad x \in \mathbb{R}, \quad x > 0.$$

Find

(a) $fg(x)$, [1]

(b) $f^{-1}(x)$, stating clearly its domain. [4]

- 3** Show that $x^2 - 2x + 7$ is always positive for all real values of x . [2]

Hence solve the inequality

$$\frac{x^2 - 2x + 7}{x^2 + 4x + 3} \geq 0. \quad [3]$$

- 4** The equation of a curve C is $y = f(x)$, where

$$f(x) = \frac{20}{1 - x^2}.$$

(i) Write down all the equations of the asymptotes of C . [2]

(ii) Sketch C , stating the equations of all the asymptotes and coordinates of the turning point. [2]

Hence find the range of values of k for which

$$(x^2 - k)(x^2 - 1) - 20 = 0$$

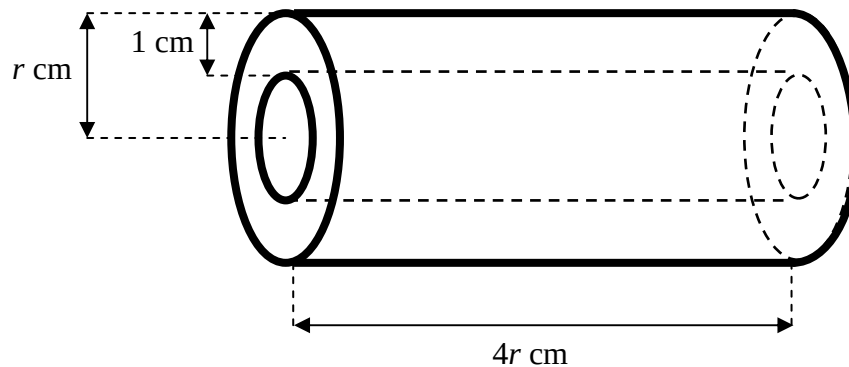
has exactly two solutions. [2]

(iii) Sketch $y = |f(x)|$. [2]

- 5 The curve C is defined by the equation $y = \frac{1}{x-1} + 2$. L_1 is the tangent to the curve at the point where $x = 0$. L_2 is another tangent to C parallel to L_1 .

- (i) Find the equations for both L_1 and L_2 . [4]
 (ii) Sketch C , L_1 and L_2 , labelling any asymptotes and axial intercepts clearly. [2]
 (iii) Find the area of the region R enclosed by C , L_2 and the line $x = 8$. [3]

6 (a)



The pipe in the diagram above is expanding in such a way that at time t seconds, its outer radius and length are given by r cm and $4r$ cm respectively. Its thickness is maintained constantly at 1 cm.

Show that the volume of the pipe is given by $V = 4\pi r(2r - 1)$. [1]

When $r = 7$, the volume of the pipe is increasing at the rate of $36 \text{ cm}^3 \text{ s}^{-1}$.

Find the rate of increase of the outer radius at the same instant. [3]

- (b) (i) Given that $y = 5\ln(2x+3) + 2x^2$, find $\frac{dy}{dx}$. [2]

- (ii) The tangent to the curve $y = 5\ln(2x+3) + 2x^2$ at $x = a$, where $a > 0$, is perpendicular to the straight line $x + 6y = 2$.

Determine the value of a . [3]

[Volume of cylinder with radius r and height $h = \pi r^2 h$]

Section B: Statistics [60 marks]

- 7** A school has 80 classes, each consisting of 25 students. 60% of students in the school subscribe to ‘Starhug’ mobile service.

In a class of 25 students, find the probability that more than half of the class subscribe to ‘Starhug’ mobile service. [2]

Using a suitable approximation, find the probability that there are less than 60 classes with more than half of the class subscribing to ‘Starhug’ mobile service. [3]

- 8** A soft drink ‘Popsi’ is sold in bottles which the manufacturer claims contain 330 ml of ‘Popsi’, on average. There are complaints from consumers that the manufacturer is overstating the amount of ‘Popsi’ in these bottles.

A consumer association takes a random sample of 1,000 bottles and the amount, x ml, of ‘Popsi’ in each bottle is determined. It is found that

$$\sum (x - 300) = 28765 \text{ and } \sum (x - 300)^2 = 1345678.$$

Find unbiased estimates of the mean and variance of the amount of ‘Popsi’ in each bottle. [2]

Test, at 5% level of significance, whether the manufacturer is overstating the amount of ‘Popsi’ in each bottle. [4]

- 9** A factory manufactures rubber balls with radius which follows a normal distribution with mean μ and standard deviation σ . It is known that 90% of them have radius at most 10 cm and 75% of them have radius at least 7 cm.

Show that $\mu = 8.03$ and $\sigma = 1.53$ approximately. [4]

The factory also produces metal pipes, each with inner diameter D cm, which would allow the manufactured balls to pass through.

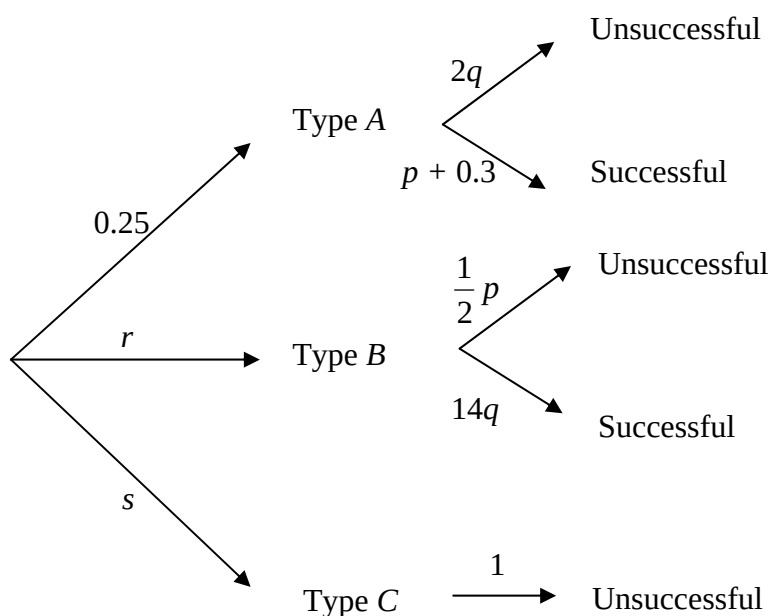
Determine the smallest integer value for D that will allow at least 90% of the balls to pass through. [4]

- 10 An employment agency receives job applications from three different background types:

Type	Applicant's background
A	At least tertiary education
B	Acquired secondary education but no lesser
C	Acquired at most primary education

0.25 of its applicants are of Type A, r are of Type B and s are of Type C, where r and s are probabilities.

The probabilities of the agency successfully job-matching those from Type A and B are given in the tree diagram below. Due to its industry specialisation, the agency would not be able to successfully job-match those from Type C.



- (i) Show that $p = 0.60$ and $q = 0.05$. [1]
- (ii) The probability that an applicant is of Type B and cannot be successfully job-matched is 0.135. Show that r and s are 0.45 and 0.30 respectively. [2]

From records, the probabilities that a successful Type A applicant and a successful Type B applicant remain in their new jobs for at least six months are 0.75 and 0.60 respectively.

- (iii) Extend the tree diagram to include this new information. [2]
Find the probability that a fresh applicant will remain in his job for at least six months. [2]
- (iv) Given that an applicant is successfully job-matched, find the probability that he will remain in his job for at least six months. [3]

- 11** A survey is conducted to study monthly expenditures of households. A sample size of 1000 households is chosen.

- (i) The table below shows the percentage of residents staying in the various types of housing in Singapore.

Home residence	Percentage
Private Housing	4.3%
Private Flat	8.1%
Public (HDB) Flats	87.6%

Explain why stratified sampling is a suitable method to select the required number of households and describe how it can be conducted. [2]

- (ii) It is found that the monthly household expenditures for the population is as follows:

Group	Housing type	Mean monthly expenditure	Standard Deviation
A	Private Housing	\$6,958	\$1,322
B	Private Flat	\$5,846	\$1,068
C	Public (HDB) Flats	\$2,804	\$355

A sample of 60 households in Group A and another sample of 80 households in Group B are randomly selected. Find the probability that the average monthly expenditure of the 60 households in Group A is at least \$1,000 more than that of the 80 households in Group B. [4]

- (iii) A large sample size of n households is randomly selected from Group C. Write down the distribution for the total monthly expenditure for n households. [1]

Find the least value of n such that the probability of the total monthly expenditure of n households being more than \$200,000 is at least 0.95. [3]

- 12** A factory produces bars of chocolate with the mass of each bar assumed to follow a normal distribution with mean 104 g and standard deviation 3 g. A bar is considered “substandard” if its mass is below 100 g. All the bars of chocolate produced will then be packed into boxes of 30 before they are delivered for sale.

- (i) Find the probability that a box contains at most 1 “substandard” chocolate bar. [3]
- (ii) Using a suitable approximation, find the probability that in 100 boxes of chocolates, that there are at least 30 boxes with at most 1 “substandard” chocolate bar in each box. [3]
- (iii) Using a suitable approximation, find the smallest value of m such that the probability of having at most m “substandard” chocolate bars in the 100 boxes exceeds 0.95. [4]

- 13** For twelve consecutive months, a factory manager recorded the number of items produced by the factory and the total cost of production. The following table summarizes the data.

Number of items produced, x (in thousands)	18	45	69	33	59	79	10	19	53
Production cost, y (in thousand \$)	37	63	84	49	79	98	32	96	71

- (i) Draw the scatter diagram for the data. [1]

Giving a reason, identify a data pair which should be regarded as suspect. [2]

Revise the data by removing the suspect data pair.

- (ii) Find the regression line of Y on X in the form $y = a + bx$. [1]

- (iii) Determine the product moment correlation coefficient. [1]

Hence comment on the reliability of your answer in (ii). [1]

The selling price of each item produced is \$1.60.

- (iv) Assuming that all items produced are sold,

- (a) find the estimated production cost for 53,000 items. [1]

Hence determine the corresponding profit. [2]

- (b) find the least number of items produced which would result in a profit. [2]

End of Paper